

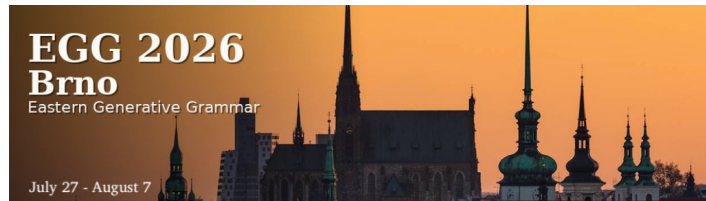


Minimalist Foundations

Kerstin Hoge

University of Oxford

<https://kxhoge.github.io/resources.html>



3 August– 7 August 2026

Course outline

Monday	From X-bar theory to Bare Phrase Structure: The emergence of Merge
Tuesday	Merge, Move, and Agree
Wednesday	The expansion of Merge: External / Internal Merge and Phase theory
Thursday	The Strong Minimalist Thesis, Merge, and the problem of labelling
Friday	The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed: Can syntax be reduced to Merge?

Outline

1. Successive cyclicity
2. The Miracle Creed framework
3. Course summary

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

- *Try your hand at some syntactic analysis*
- The phase-theoretic analysis of a *wh*-question like (1) postulates that the *wh*-phrase must move successive-cyclically through the phase edges, first to SpecvP and then to SpecCP.

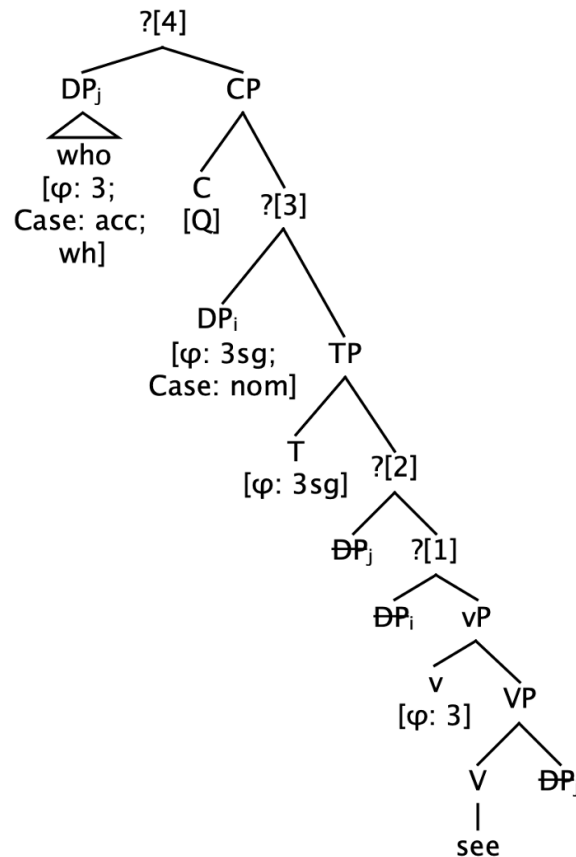
(1) Who did Mary see?

Consider the individual steps in the derivation of (1). Do the resulting syntactic objects pose labelling problems, and if so, how does the computational system resolve them?

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

(2)

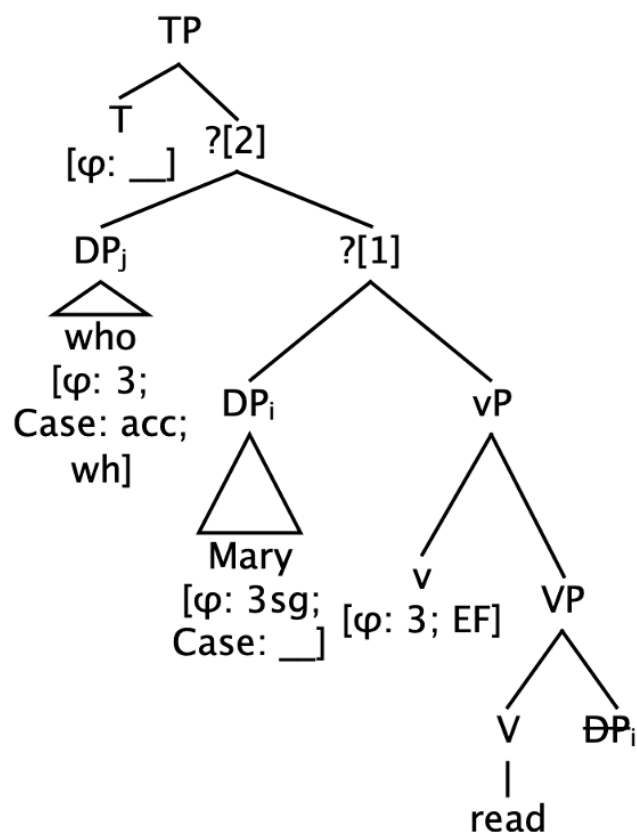


- Problems of labelling arise at $?[1-4]$
- What, if anything, forces Internal Merge($DP_j, \{DP_i, vP\}$)?
- What forces Internal Merge(DP_j, CP)?
- If T probes for a goal with matching phi-features, why is Agree not established with (structurally closer) DP_j ?

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

(3)



- What, if anything, forces Internal Merge(DP_j , $\{DP_i, vP\}$)?
- Given the PIC, the wh-object must raise to the edge of the vP phase in order to remain accessible for subsequent Internal Merge at the edge of the next phase, CP.
- Chomsky (2008) treats Merge as freely available, including Internal Merge, but still assumes that particular applications of Internal Merge (most notably movement to the edge of a phase) are made available by a so-called **edge feature** (EF) on the relevant phase head.

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

- ‘For an LI to be able to enter into a computation, merging with some SO, it must have some property permitting this operation. A property of an LI is called a feature, so an LI has a feature that permits it to be merged. Call this the **edge feature** (EF) of the LI’ (Chomsky 2008: 139).
- ‘If only phase heads trigger operations (as I will assume), then IM will satisfy EF only for phase heads’ (Chomsky 2008: 144).

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

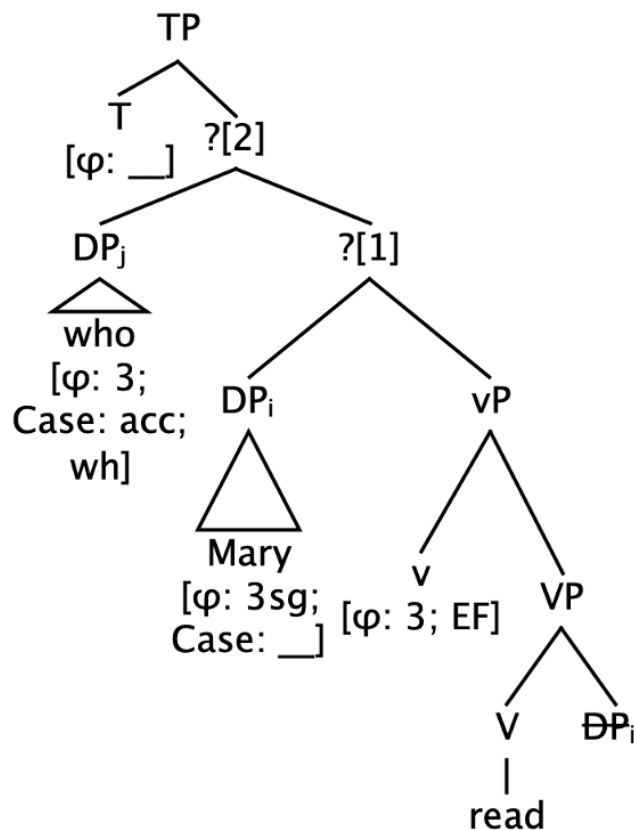
- Note that from Chomsky (2013) onwards, edge features are no longer assumed to be part of C_{HL} .

‘In earlier years it was assumed generally (by me in particular) that displacement is problematic, an “imperfection” of language, to be explained by external factors, perhaps parsing considerations. A residue of that error is the belief that EM is somehow simpler and preferable to IM. There is no basis for that belief. If anything, IM is simpler, since it requires vastly less search than EM (which must access the workspace of already generated objects and the lexicon). But the question doesn’t arise: *both are freely available*, given the simplest combinatorial operation, Merge’ (Chomsky 2013: 41).

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

(4)



- If T probes for a goal with matching phi-features, why is Agree not established with (structurally closer) DP_j?

- Since DP_j enters into Agree with v, its Case feature is valued as accusative. Once the Case feature of DP_j has been valued, DP_j becomes “inactive” and can no longer enter into an Agree relation with a higher probe head. This allows T to establish Agree with a structurally more distant active DP, viz. the external argument *Mary*.

The Miracle Creed: Can syntax be reduced to Merge?

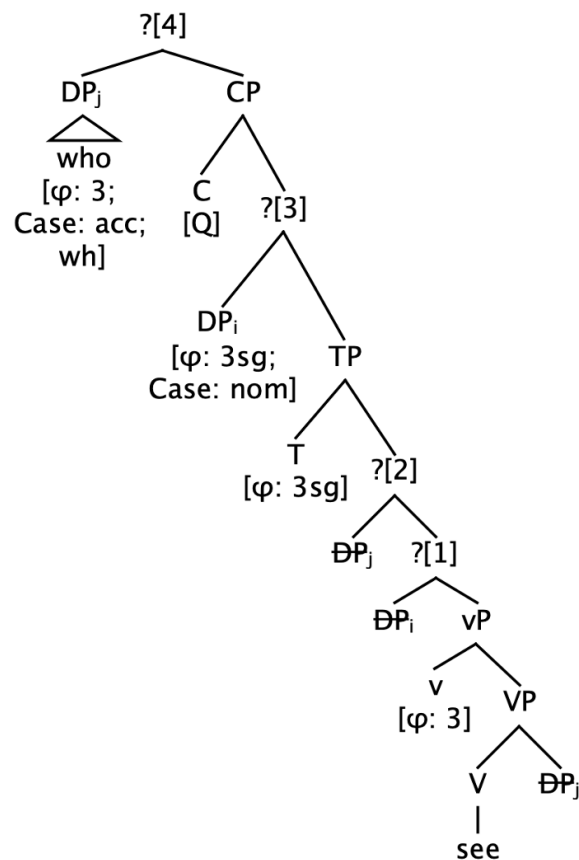
Successive Cyclicity

‘If uninterpretable features serve to implement operations, we expect that it is structural Case that enables the closest goal G to select P(G) [= the phrase containing the goal] to satisfy the EPP by Merge. Thus, if structural Case has already been checked (deleted), the phrase P(G) is “frozen in place”, unable to move further to satisfy the EPP in a higher position. More generally, uninterpretable features render the goal *active*, able to implement an operation: to select a phrase for Merge (pied-piping) or to delete the probe. The operations Agree and Move require a goal that is both local and active’
Chomsky 2000: 123).

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

(5)



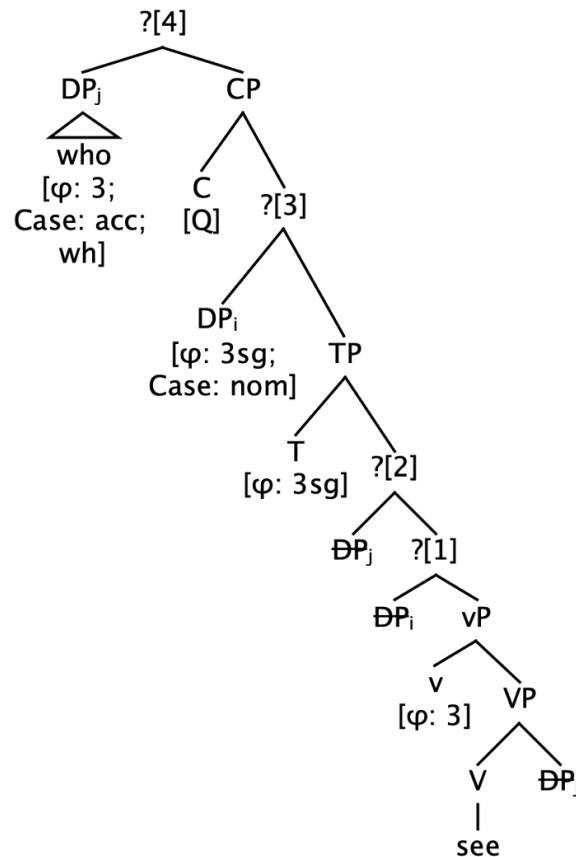
- What forces Internal Merge(DP_j , CP)?

- If Internal Merge is freely available, the displacement of the wh-phrase to clause-initial position need not be triggered by a dedicated movement feature, such as an edge feature.

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

(6)



• Problems of labelling arise at $\{ \varphi: 3 \}$

• Internal Merge(DP_i, TP) modifies the problematic configuration {DP_i, vP}, allowing $\{ \varphi: 3 \}$ to be labelled as vP.

• Internal Merge(DP_j, CP) modifies the problematic configuration {DP_j, {DP_i, vP}}. allowing $\{ \varphi: 2 \}$ to be labelled as vP.

• Since DP_i and T enter into an agreement relation and share ϕ -features, the two members of the SO {DP_i, TP} can be identified by their shared ϕ -feature set. The resulting SO ($\{ \varphi: 3 \}$) can therefore be labelled by ϕ .

• For SO {DP_j, CP}, the labelling problem is likewise resolved through feature sharing.

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

‘Let’s turn next to (B), {XP, YP} in situ without raising ... How then can it be labeled?

The solution may lie in the fact that the most prominent feature of α and of β is shared, namely the interrogative feature Q, a feature of C and the head of α if we adopt a plausible analysis of interrogative wh-phrases that takes Q to be the most prominent element (Cable, 2007, 2010; Narita, 2011). Searching {XP, YP}, then, LA finds the same most prominent element -- Q -- in both terms, and can take that to be the label of α ’ (Chomsky 2013: 45).

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

- Long-distance displacement of a wh-phrase exemplifies both strategies for resolving the labelling problem, (A) and (B), at different stages of the derivation.

(7) Who did Alex [_{VP} ~~who~~ claim [_{CP} ~~who~~ that Mary [_{VP} who [_{VP} saw ~~who~~]]]]?

‘The interesting case is $SO = \{XP, YP\}$, neither a head ... Here minimal search is ambiguous, locating the heads X, Y of XP, YP, respectively. There are, then, two ways in which SO can be labeled: (A) modify SO so that there is only one visible head, or (B) X and Y are identical in a relevant respect, providing the same label, which can be taken as the label of the SO’
(Chomsky 2013: 43)

The Miracle Creed: Can syntax be reduced to Merge?

Successive Cyclicity

‘The most general case of lack of label is successive-cyclic movement. The intermediate steps are of the form $\{\alpha \text{ XP, YP}\}$, where XP can be for example a wh-phrase with YP a CP. The syntactic object α cannot be labeled, but it must be interpreted, if only for theta-marking. If XP raises, then α will be labeled Y, as required. Therefore XP must raise, and successive cyclic movement is forced. We see here again the empirical advantages of dropping the requirement that every syntactic object must be labeled, as in the more complex head-oriented concept of Merge in X-bar theory and its descendants’
(Chomsky 2013: 44).

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- Building on the conception of Merge and labelling developed in Chomsky (2013, 2015), Chomsky (2023, 2024) develops a more radically minimalist approach to narrow syntax. The Miracle Creed framework retains the assumption that Merge, both External and Internal, is freely available, while constraining its application by ‘Principle T’, according to which ‘all relations and structure-building operations (SBO) are thought-related, with semantic properties interpreted at CI’ (Chomsky 2024: 5).

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- The Miracle Creed framework derives its name from the metaphor invoked by Chomsky to describe his conviction that the apparent complexity of language can emerge from remarkably simple assumptions about C_{HL} .

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

‘... there’s a thesis called the Strong Minimalist Thesis, which proposes that the internal system is basically perfect (Chomsky 2000). It’s designed optimally using the simplest computational operation. Why should we think of accepting it? The reason is what’s been discovered in science in the last couple of hundred years, what Albert Einstein called the miracle creed. Quoting Einstein, the true scientist believes that ‘the totality of sensory experience can be “comprehended” on the basis of a conceptual system built on premises of great simplicity. The skeptic will say that this is a “miracle creed”. Admittedly so, but it is a miracle creed which has been borne out to an amazing extent by the development of science’ (Einstein 1950: 13).

In fact, by now, it’s simply taken for granted by scientists. Your goal is to show that the miracle creed has been achieved. If it’s good enough for all of science, it’s good enough for us. *Maybe language falls together with everything else in the world and satisfies the miracle creed*, the internal system at least (and probably, in different ways, externalization too). If that’s true, we have the ultimate in explanation for this fundamental property of language’ (Chomsky 2023: 8).

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- In what sense are Merge and the syntactic objects it constructs ‘thought-related’, as required by Principle T?

Several categories of thought are relevant to language structure and use. One category is *propositional*: basic theta-structure. A second is *clausal*: force- and information-related (interrogative, topic, focus,...). The familiar property of **duality of semantics** (Chomsky 2024: 4-5).

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- ‘If you look quite generally at the interpretation of expressions, it falls into two categories. There is **one category which yields argument structure** (theta roles and the interpretation of complements of functional elements). There is **another category which is involved in displacement**, which has kind of discourse-oriented or information-related properties or scopal properties and so on, but not argument properties. That’s **duality of semantics**. If you think about it a little further, you see that the first type, argument structure is invariably given by external MERGE. The second type, non-argument structure (other factors) is always given by internal MERGE.’
(Chomsky 2020: 44)

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- In the Miracle Creed framework,
 - **External Merge** constructs theta-structure by establishing predicate-argument relations;
 - **Internal Merge** applies to already theta-marked syntactic objects and remerges them in non-theta positions, where they receive interpretations relating to discourse, scope, or other properties of the CI interface.

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- If Internal Merge applies only to theta-marked SOs, then a SO occupying a non-theta position is no longer eligible for further Internal Merge. It follows that a SO cannot undergo successive Internal Merge from one non-theta position to another, thereby **ruling out successive-cyclic movement in the Miracle Creed framework**.

- (8) a. $*[_{CP} \text{who } C_Q [_{TP} \text{Mary } [_{VP} \text{who } [_{VP} \text{see who}]]]]$
b. $[_{CP} C_Q [_{TP} \text{Mary } [_{VP} \text{who } [_{VP} \text{see who}]]]]$

- How, then, can the sentence *Who did Mary see?*, in which *who* has scope over the entire clause, be derived?

The Miracle Creed: Can syntax be reduced to Merge?

The Miracle Creed framework

- Chomsky (2024) proposes that **C_Q accesses the occurrence of *who* in the vP phase** to obtain the ‘instructions’ required by the interfaces: at the CI interface, *who* is interpreted as having clause-wide scope; and at the SM interface, *who* is pronounced in clause-initial position.

(9) [CP C_Q [TP Mary [vP who [vP see ~~who~~]]]]



‘For ease of exposition, we can think of the element E that is IM-ed to the phase edge as being put in a box, separate from the ongoing derivation D. E must however be accessible to D at later phase levels for interpretation at the interfaces. [Chomsky (2021)] maintained the standard assumption that E moves phase-by-phase to SPEC-phase, where it is interpreted at CI (also at SM if Externalization is activated). Now there’s no movement and the boxed element E is accessible at the phase level’ (Chomsky 2024: 8).

The Miracle Creed: Can syntax be reduced to Merge?

Course summary

- How have successive phases of Minimalist theorising progressively redefined the role of Merge?

Publication	Syntactic operations	Relation of Merge and labelling
1993/1995	Merge (free); Move (costly)	Merge creates inherently labelled SOs
2000	Merge (free); Move (costly) = Merge + Agree + Pied-Piping	Merge creates inherently labelled SOs
2004	Merge: External (free) & Internal (free but constrained by PIC); Agree	Merge creates inherently labelled SOs
2013	Merge: External (free) & Internal (free but constrained by PIC); Agree	separation of Merge and labelling (now delegated to a separate Labelling Algorithm)
2024	Merge: External (free but permitted only if it builds theta-structure) & Internal (free but permitted only if it applies to a theta-marked SO); Agree	separation of Merge and labelling (now delegated to a separate Labelling Algorithm)



Thank you!

The Miracle Creed: Can syntax be reduced to Merge?

References

- Chomsky, Noam. 1993. A Minimalist program for linguistic theory. In Kenneth Hale & Samuel J. Keyser (eds.), *The view from Building 20: Essays in linguistics in honor of Sylvain Bromberger*, 1–52. Cambridge, MA: MIT Press.
- Chomsky, Noam. 1995. *The Minimalist program*. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In Roger Martin, David Michaels & Juan Uriagereka (eds.), *Step by step: Essays on Minimalist syntax in honor of Howard Lasnik*, 89–155. Cambridge, MA: MIT Press.
- Chomsky, Noam. 2004. Beyond explanatory adequacy. In Adriana Belletti (ed.), *Structures and beyond: The cartography of syntactic structures*, vol. 3, 104–131. New York & Oxford: Oxford University Press.
- Chomsky, Noam. 2008. On phases. In Robert Freidin, Carlos P. Otero & Maria Luisa Zubizarreta (eds.), *Foundational issues in linguistic theory: Essays in honor of Jean-Roger Vergnaud*, 133–166. MIT Press. doi:10.7551/mitpress/7713.003.0009. <https://direct.mit.edu/books/book/3235/chapter/95389/On-Phases>.
- Chomsky, Noam. 2013. Problems of projection. *Lingua* 130, 33–49. doi:10.1016/j.lingua.2012.12.003. <https://linkinghub.elsevier.com/retrieve/pii/S002438411200277X>.
- Chomsky, Noam. 2015. Problems of projection: Extensions. *Linguistik Aktuell/Linguistics Today* 223. Amsterdam & Philadelphia, PA: John Benjamins, 3–16. doi:10.1075/la.223.01cho. <https://benjamins.com/catalog/la.223.01cho>.
- Chomsky, Noam. 2020. The UCLA lectures. lingbuzz/005485.
- Chomsky, Noam. 2021. Minimalism: Where are we now, and where can we hope to go. *Gengo Kenkyu (Journal of the Linguistic Society of Japan)* 160, 1–41. doi:10.11435/gengo.160.0_1. https://doi.org/10.11435/gengo.160.0_1.
- Chomsky, Noam. 2023. Language and the Miracle Creed. *Researching and Teaching Chinese as a Foreign Language* 4(1), 1–12. doi:10.1558/rtcfl.25349. <https://utppublishing.com/doi/10.1558/rtcfl.25349>.
- Chomsky, Noam. 2024. The Miracle Creed and SMT. In Matteo Greco & Davide Mocci (eds.), *A Cartesian dream: A geometrical account of syntax in honor of Andrea Moro*, 17–40. Barcelona: LingBuzz Press.